**3.**

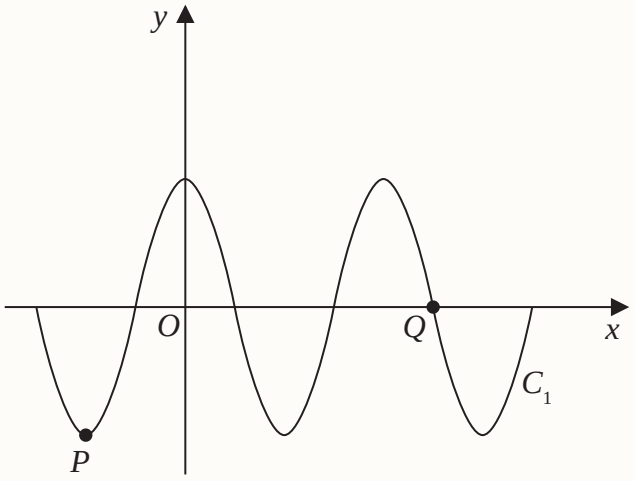



**Figure 1**

Figure 1 shows a sketch of part of the curve $C_1$ with equation $y = 4\cos x°$

The point $P$ and the point $Q$ lie on $C_1$ and are shown in Figure 1.

(a) State

(i) the coordinates of $P$,

(ii) the coordinates of $Q$.

**(3)**

The curve $C_2$ has equation $y = 4\cos x° + k$, where $k$ is a constant.

Curve $C_2$ has a minimum $y$ value of $-1$

The point $R$ is the maximum point on $C_2$ with the smallest positive $x$ coordinate.

(b) State the coordinates of $R$.

**(2)**